



Health & Stress:

What is stress? What is the stress response?

By the end of class...

- Define stress and stressors.
- Differentiate types of stressors and stress responses.
- Explain how people appraise stress.
- Describe the effects of stressors on mental and physical health.
- Explain how stress contributes to disease.
- Identify strategies to build resilience.





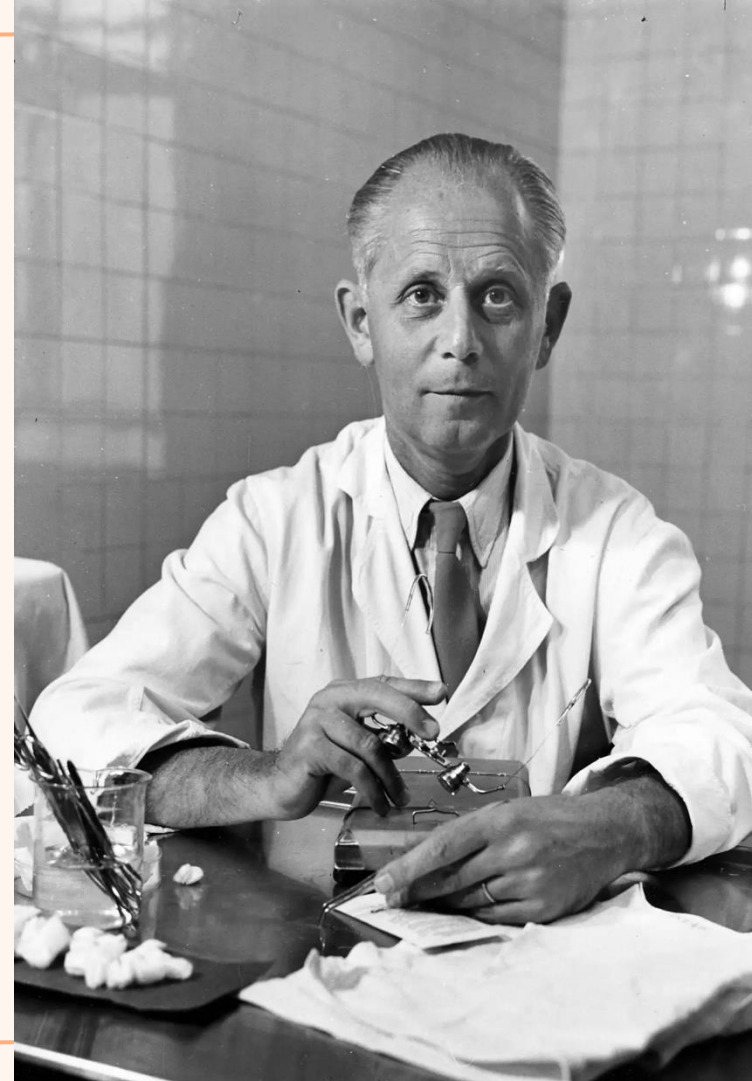
What Is Stress?

- **Stress** – the physical and psychological responses to internal or external stressors
- **Stressors** – internal or external circumstances and/or stimuli perceived as a threat, challenge or deviation from a balanced state



Good versus Bad Stress

- Hans Selye → the “Father of Stress Research” and coined the term *stress*
- Types of stressors:
 - **Eustress** – a reaction to **good** stress
 - **Distress** - a reaction to **bad** stress



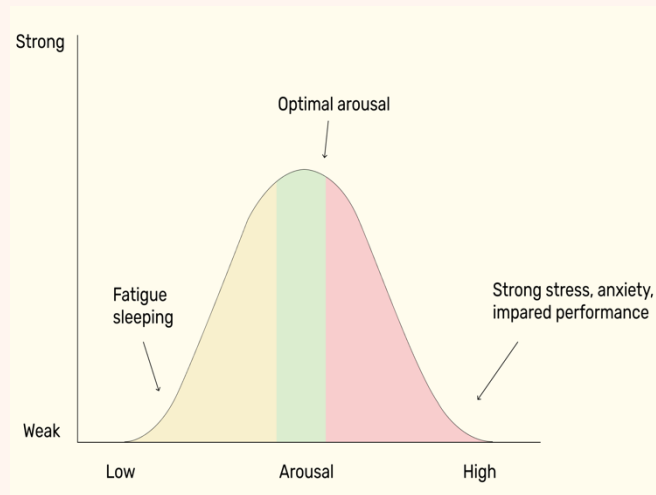
How Much Stress is Optimal?

Yerkes-Dodson Law

- Too little stress → underperformance
- Moderate stress → peak performance
- Too much stress → peak performance declines

Maximal Adaptability Model

- Challenges the Yerkes-Dodson Law
- Humans are highly adaptable to stress
- High performance can be maintained under both underload and overload



Yerkes-Dodson Law

Stressors and the Stress Response

Types of stressors can be defined by duration and severity

Stressor	Duration	Severity	Examples?
Acute Stressors	Short-term circumstances, lasting minutes to hours	Varies from mild to moderate to severe	Childbirth Divorce Taking an exam Public speech
Chronic Stressors	Enduring circumstances, lasting weeks to years	Varies from mild to moderate to severe	Living in poverty
Traumatic Stressors	Immediate threat to physical integrity	Most severe stressors	Child abuse Natural disasters

The Stress Response: An Evolutionary Perspective

Stress responses - integrated psychological and biological responses to stressors that help restore a balanced state

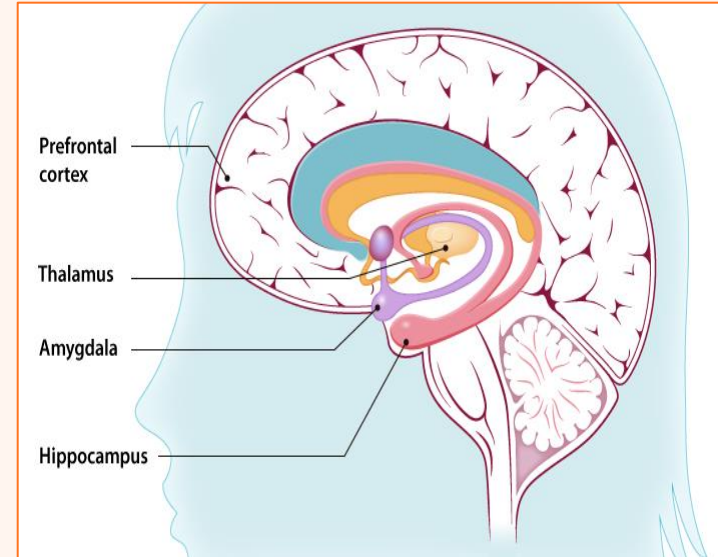
- Highly coordinated processes
- Protect the body from harm
- Mobilize energy to cope with stress, prevent injury, and reduce infection risk

Homeostasis – the body's balanced state

- Stressors disrupt homeostasis; the stress response works to restore it

Neurobiological Responses to Stressors

- Earliest stress responses typically occur in the **amygdala**
 - Referred to as the “*neural watchdog*”
- The **hippocampus** is central to learning and memory
- The **prefrontal cortex** works to regulate the amygdala responses to potential stressors

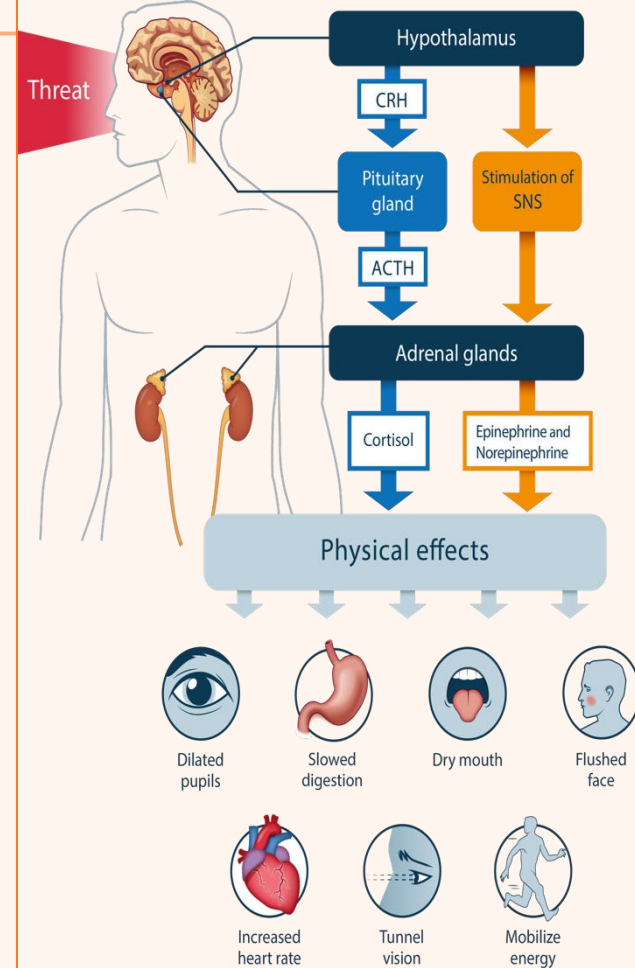


Stress responses occur in the amygdala, hippocampus, and prefrontal cortex.

Hormonal Responses to Stressors

The brain registers a stressor and rapidly activates two biological systems:

1. The **autonomic nervous system (ANS)**
 - Includes the **parasympathetic nervous system (PNS)** and the **sympathetic nervous system (SNS)**
2. The **hypothalamic pituitary adrenal (HPA) axis**
 - Release catecholamines and cortisol to help body prepare for stressor



Stress response in the ANS and the HPA axis.

Immune System Responses to Stressors

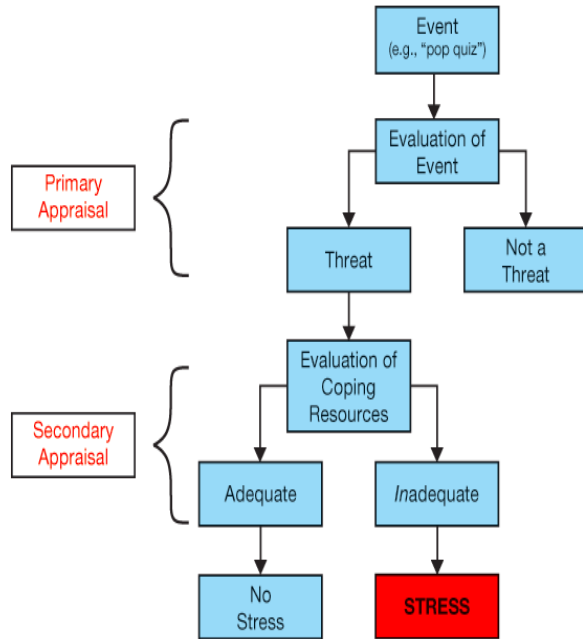
How stress responses enhanced the survival and success of our ancestors

Everyday, acute stressors activate the inflammatory response of the immune system

Activation of inflammatory response involves increased levels of proteins that promote inflammation

Inflammatory response allows body to cope with increased risk of infection and injury from stressors

Perception Matters: Stress and Coping Theory

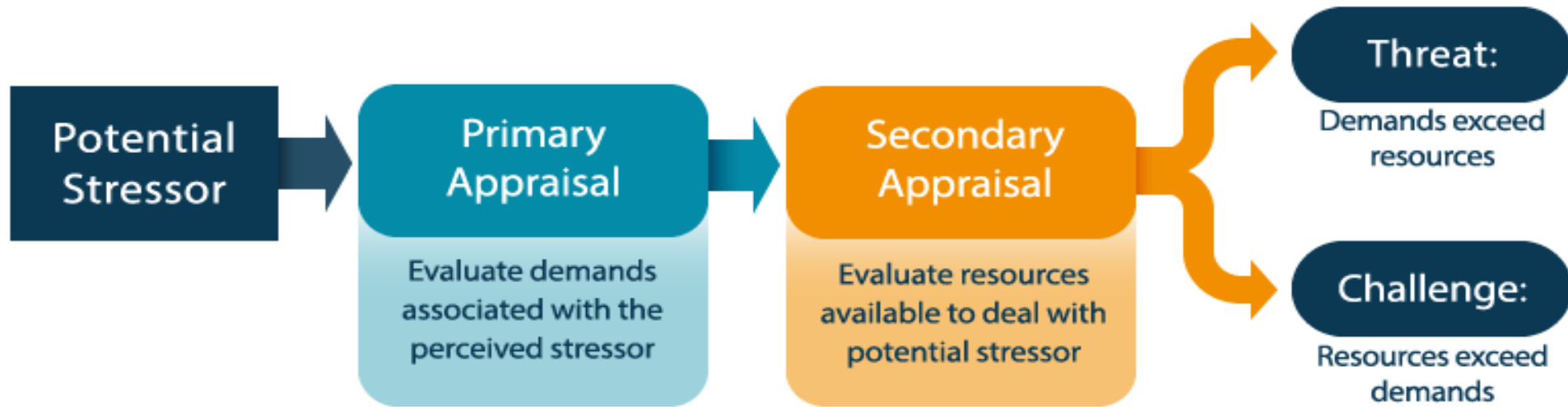


The role of primary and secondary appraisals in determining whether an event is a stressor

A big source of individual differences in stress response is how we **appraise** stressors

- **Primary appraisal** → determining whether a stimulus is a threat
- **Secondary appraisal** → determining whether you can handle the stressor
 - 1) Do you have control over it?
 - 2) Do you have the coping mechanisms to deal with it?
 - If so, it causes less stress
 - If not, it's a stressful event

Perception Matters: Challenge and Threat Theory

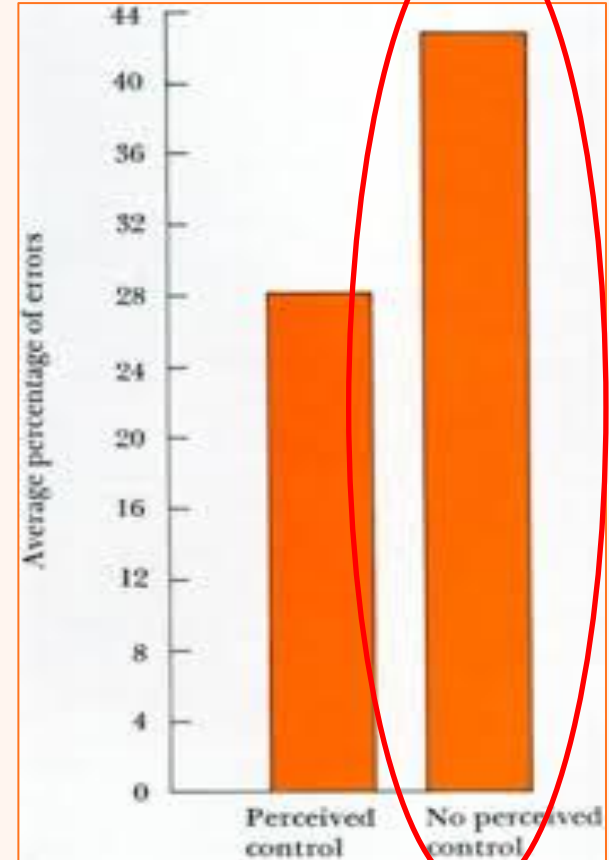


Challenge and Threat Theory:

A flowchart shows the evaluation of a potential stressor as a threat or challenge

Perception Matters: Glass & Singer (1972)

- Glass & Singer (1972) placed individuals into a room and asked them to complete difficult puzzles
 - No perceived control: Unpredictable loud noise
 - Perceived control: Could turn the noise off (but didn't)
 - The 'no perceived control' group performed worse
- **Take-home:** Perceived control shapes the stress response



Question Time!



Which statement BEST reflects secondary appraisal according to Stress and Coping Theory?

- A. This exam is worth 40% of my grade
- B. This situation could seriously affect my future
- C. I have enough time, support, and study strategies to prepare for this exam
- D. Public speaking makes me uncomfortable

Question Time!



Two students complete the same difficult math test while construction noise occurs outside. Student A has no control over the noise. Student B is told they can close a soundproof window at any time, although they never actually do.

Based on Glass & Singer (1972), who is likely to perform better?

- A. Student A
- B. Student B
- C. Neither
- D. Both student have an equal chance of performing well

Stress and Mental Health

- Psychological stressors are widely recognized as major risk factors for common psychological disorders (e.g., Post-Traumatic Stress Disorder (PTSD))
- Stressors involving **social rejection** are most likely to trigger depression
- Loss events, humiliation events, entrapment situations, and danger events are strong risk factors for psychological disorders
- **Childhood stressors** particularly harmful with long-lasting effects



Stress and Physical Health

Acute Illnesses

- Stress impacts risk for development (e.g., colds)
- **Asthma attacks** can be more severe when stressed
- **Latent Viruses** can be reactivated under stress (e.g., Herpes simplex virus)

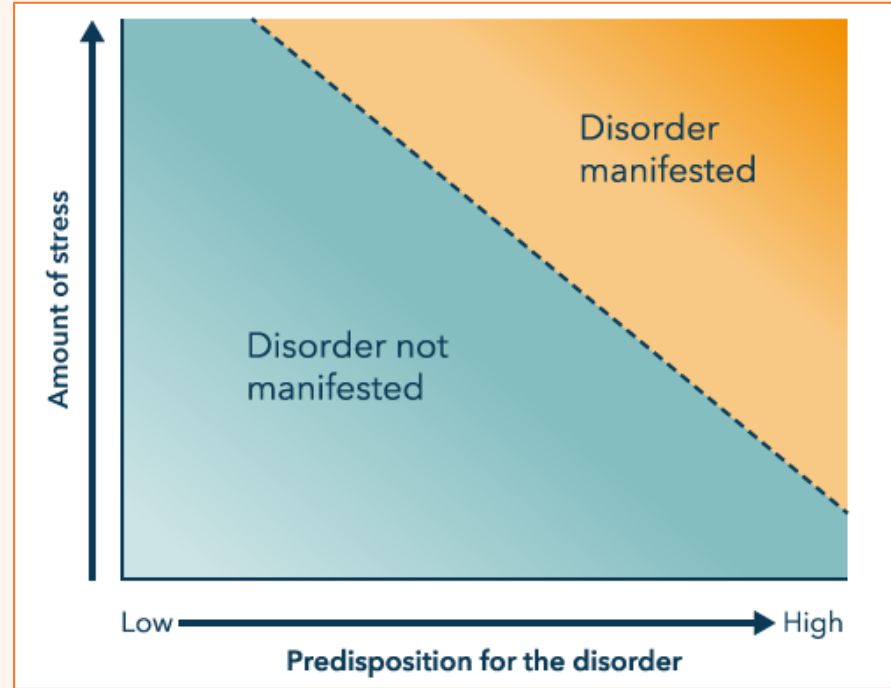


Chronic Illnesses

- More severe when stressed
- **Cardiovascular Diseases** influenced by stress (acute and chronic stress)
- **Autoimmune disorders** influenced by stress (e.g., Multiple sclerosis, lupus, Chron's disease)

The Diathesis-Stress Model

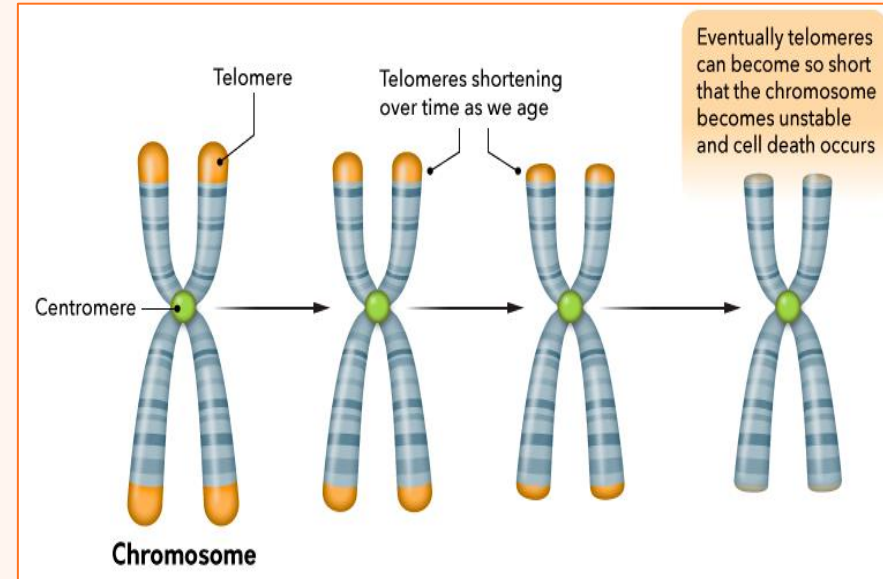
- **Diathesis-Stress Model:** “Theory that mental and physical disorders develop from a genetic or biological predisposition for that illness (diathesis) combined with stressful conditions that play a participating or facilitating role.”
(*APA Dictionary*)
- Plasticity versus vulnerability factor



The Diathesis-Stress Model

Pathways from Stress to Mental and Physical Disease

- **Chronic Inflammation**
 - Prolonged exposure to inflammatory proteins can lead to cell death and tissue damage
- **Biological Aging**
 - Older cells unable to divide or function
 - Increased risk for diseases of aging (e.g., cardiovascular disease, autoimmune disorders)

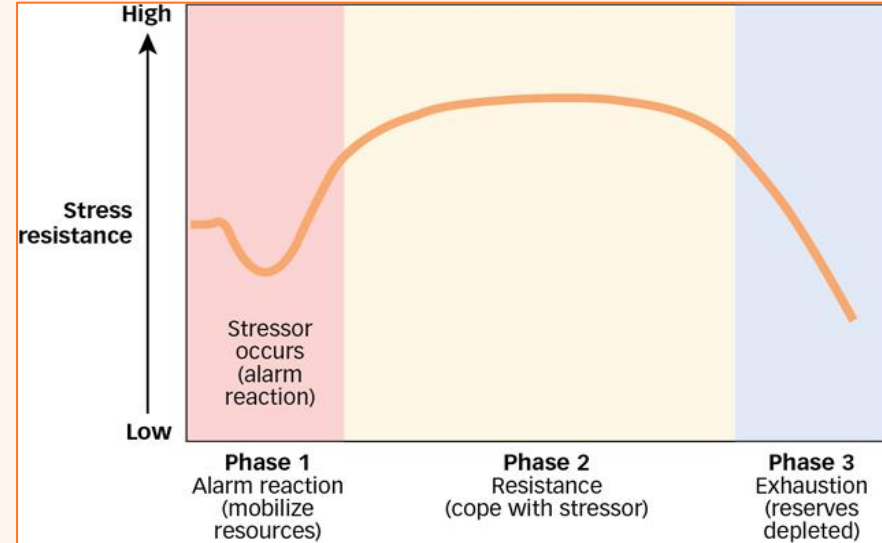


Telomeres are DNA-protein complexes that cap chromosomes and protect against damage to DNA

General Adaptation Syndrome

Stages of General Adaptation Syndrome

1. **Alarm Phase** – initial, healthy reaction to stress
2. **Resistance Phase** – body adapts to high stress; non-stress-related processes are shut down
3. **Exhaustion Phase** – body cannot cope with other processes being shut down; illness, injury or death can occur



The three phases of the General Adaptation Syndrome



Reducing Stress & Enhancing Resilience

- Behaviours you can engage in to protect yourself from the negative effects of stress
- Can have direct effect on health outcomes and stress-buffering effects

- Behaviours to improve resilience:
 1. Physically active lifestyle
 2. Healthy sleep schedule
 3. Maintain a healthy diet
 4. Maintain social support
 5. Use relaxation techniques



Question Time!



Emma and Olivia both lose their jobs unexpectedly. Emma has a family history of depression and has experienced several adverse childhood events. Olivia has no known family history of mental illness and a strong social support network.

According to the Diathesis-Stress Model, who is at greater risk of developing depression?

- A. Emma, because stress interacts with an underlying predisposition
- B. Olivia, because losing a job is a severe stressor
- C. Emma, because genetics alone determine whether someone develops depression
- D. Olivia, because the stressor alone determines risk

You should now be able to...

- Define stress and stressors. ✓
- Differentiate types of stressors and stress responses. ✓
- Explain how people appraise stress. ✓
- Describe the effects of stressors on mental and physical health. ✓
- Explain how stress contributes to disease. ✓
- Identify strategies to build resilience. ✓

